



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

**DATA STRUCTURES AND ALGORITHMS 3
(25CS2103E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Section: 1	Team No: 21	Project Title: Interactive Complexity Theory Learning Platform
------------	-------------	--

Abstract

This project proposes an interactive learning platform for computational complexity theory, designed to help users understand the fundamental concepts of P, NP, NP-Complete, and NP-Hard problems through practical and visual learning. The platform combines theoretical explanations with interactive examples, graph visualizations, algorithm demonstrations, and problem-solving activities to make complex concepts easier to understand.

Users can explore important problems such as Travelling Salesman Problem, Knapsack, Vertex Cover, and Hamiltonian Cycle, along with demonstrations of polynomial-time algorithms, reductions, and approximation techniques. The system also provides quizzes and challenges to assess conceptual understanding, while performance analysis tools allow users to compare algorithms based on execution time and input size.

A key feature of the platform is the ability to automatically evaluate user-generated problem instances, providing immediate feedback on correctness and algorithmic performance. By integrating visualization, experimentation, automated evaluation, and interactive assessment, the proposed system aims to bridge the gap between theoretical concepts and their practical implementation. The platform ultimately provides a hands-on environment for students to develop a stronger conceptual understanding of computational complexity and advanced algorithmic problem-solving.

DSA Concepts / Algorithms Used

1. Graph Algorithms – BFS, DFS, Dijkstra
2. Minimum Spanning Tree – Prim's, Kruskal's
3. Dynamic Programming – Knapsack Problem
4. Greedy & Approximation Algorithms – TSP, Vertex Cover
5. Backtracking – Hamiltonian Cycle
6. Complexity Analysis & Reductions – P, NP, NP-Complete, NP-Hard

Team Details

Roll Number	Name	Signature
2520030446	K. Satwik	
2520030610	K. Sai Krishna	
2520030530	K. Charan	

Guide Name and Signature (remarks if any)**Guide Name:****Signature:****Remarks:**